

Disproof of Conjecture 00000001289: no graph on $2^n \geq 4$ vertices has diameter $2^n - 1$ and chromatic number 2^n

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Abstract

Conjecture 00000001289 asserts that the “nim graph” $Q_{\oplus}(n)$ of the hypercube (vertices: the 2^n vertices of the n -cube; edges “given by $\oplus$ ”) has diameter $2^n - 1$ and chromatic number 2^n . The edge rule is not specified, but this does not matter: we show that *no* simple graph on $N \geq 3$ vertices has chromatic number N and diameter $N - 1$, because chromatic number N forces the graph to be complete, and a complete graph has diameter 1. Hence the conjecture fails for every $n \geq 2$, already at $n = 2$, whatever symmetric (undirected) edge set is meant. For the two natural readings we also compute the true values: the hypercube Q_n ($u \sim v$ iff $u \oplus v$ is a power of two) has $\chi = 2$ and diameter n ; the complete reading ($u \sim v$ iff $u \oplus v \neq 0$) has $\chi = 2^n$ and diameter 1. Everything is formalized in Lean 4 (core library only).

1 The conjecture

Definition: The nim graph of the hypercube (edges given by $\oplus$). *Conjecture:* The diameter and chromatic number of the nim graph $Q_{\oplus}(n)$ are $2^n - 1$ and 2^n respectively (nim graph parameters).

The Chinese version says the same. We write $N = 2^n$. The vertex set of $Q_{\oplus}(n)$ is the vertex set $\{0, 1\}^n$ of the n -cube; we identify a vertex with the integer $\{0, \dots, N - 1\}$ whose binary digits it lists, so that $\oplus$ is bitwise exclusive or (nim addition). The claim to be refuted is

$$(C) \quad \text{for every } n \geq 1 : \quad \text{diam } Q_{\oplus}(n) = 2^n - 1 \quad \text{and} \quad \chi(Q_{\oplus}(n)) = 2^n.$$

It is a conjunction of two equalities for each n , and a statement about all n ; to refute it, it suffices to show that the conjunction fails for one n . We show that it fails for every $n \geq 2$.

2 Definitions and readings

Definition 1. A *simple graph* G on $V = \{0, \dots, N - 1\}$ is a symmetric, loopless adjacency relation on V . A *proper c -colouring* is a map $\text{col} : V \rightarrow \{0, \dots, c - 1\}$ with $\text{col}(u) \neq \text{col}(v)$ whenever $u \sim v$. The *chromatic number* $\chi(G)$ is the least c for which a proper c -colouring exists (it exists for $c = N$). A *walk* of length k from u to v is a sequence $u = w_0, w_1, \dots, w_k = v$ with $w_{i-1} \sim w_i$; $\text{dist}(u, v)$ is the least length of such a walk (∞ if there is none) and $\text{diam } G = \max_{u,v} \text{dist}(u, v)$. A disconnected graph has infinite diameter.

What “edges given by $\oplus$ ” can mean. The conjecture does not say which pairs are adjacent. Any rule of the form “ $u \sim v$ iff $u \oplus v \in S$ ” for a set S gives the Cayley graph $\text{Cay}((\mathbb{Z}/2)^n, S)$. Two choices are natural:

(H) $S = \{1, 2, 4, \dots, 2^{n-1}\}$: $u \sim v$ iff u and v differ in exactly one bit. This is the hypercube graph Q_n itself.

(K) $S = \{1, \dots, N-1\}$: $u \sim v$ iff $u \oplus v \neq 0$. This is the complete graph K_N .

Every rule that depends only on $u \oplus v$ is symmetric, since $u \oplus v = v \oplus u$. It may be taken loopless, since $u \oplus u = 0$. A graph with a loop has no proper colouring at all, so it cannot have $\chi = 2^n$. Multiple edges change neither χ nor distances. We therefore make *no* assumption about the edge set beyond simplicity: our main result covers every simple graph on the 2^n vertices. Directed readings are discussed in Remark 8.

3 The main argument

Lemma 2. *Let G be a simple graph on N vertices with $\chi(G) = N$. Then G is complete.*

Proof. Suppose $u \neq v$ are not adjacent. Colour v with the colour of u and give the other $N-1$ vertices $N-1$ distinct colours. Explicitly, with $s(w) = w$ for $w < v$ and $s(w) = w-1$ for $w > v$, put $\text{col}(w) = s(w)$ for $w \neq v$ and $\text{col}(v) = s(u)$. Then col takes values in $\{0, \dots, N-2\}$, and s is injective on $V \setminus \{v\}$. Two vertices with the same colour are either equal or the pair $\{u, v\}$, which is not an edge. So col is a proper $(N-1)$ -colouring, and $\chi(G) \leq N-1$, a contradiction. $\square$

Theorem 3. *Let $N \geq 3$. No simple graph on N vertices has $\chi(G) = N$ and $\text{diam } G = N-1$.*

Proof. If $\chi(G) = N$ then G is complete by Lemma 2, so every two vertices are joined by a walk of length at most 1 and $\text{diam } G \leq 1 < 2 \leq N-1$. $\square$

Corollary 4. *For every $n \geq 2$ and every simple graph G on the 2^n vertices of the n -cube, $\text{diam } G = 2^n - 1$ and $\chi(G) = 2^n$ cannot both hold. In particular Conjecture 00000001289 is false, whatever symmetric (undirected) edge set “given by $\oplus$ ” is meant; the smallest counterexample is $n = 2$.*

Proof. Apply Theorem 3 with $N = 2^n \geq 4$. $\square$

For $n = 1$ ($N = 2$) the clause can hold: $Q_{\oplus}(1) = K_2$ under both readings, with diameter $1 = 2^1 - 1$ and $\chi = 2$. So the predicates are not vacuous and the conjecture first fails at $n = 2$.

4 The actual values under the natural readings

Proposition 5. *Let $n \geq 1$ and $N = 2^n$.*

(H) *The hypercube Q_n has $\chi(Q_n) = 2$ and $\text{diam } Q_n = n$.*

(K) *The complete reading K_N has $\chi(K_N) = N$ and $\text{diam } K_N = 1$.*

Proof. (H) Colour u by the parity of the number of one-bits of u . Adjacent vertices differ in exactly one bit, so their parities differ; there is an edge, so $\chi = 2$. A walk flips one bit per step, so $\text{dist}(u, v)$ is at least the Hamming distance of u and v , and flipping the differing bits one at a time achieves it. Hence $\text{dist}(u, v)$ equals the Hamming distance, which is at most n , with equality for $u = 0$, $v = N-1$.

(K) Any two distinct vertices are adjacent, so a proper colouring is injective and needs N colours (pigeonhole); the identity colouring uses N . Distinct vertices are at distance 1. $\square$

n	claimed (diam, χ)	(H) hypercube	(K) complete	any simple graph
1	(1, 2)	(1, 2)	(1, 2)	holds for K_2
2	(3, 4)	(2, 2)	(1, 4)	impossible
3	(7, 8)	(3, 2)	(1, 8)	impossible
$n \geq 2$	$(2^n - 1, 2^n)$	$(n, 2)$	$(1, 2^n)$	impossible

Under (H) both equalities fail for $n \geq 2$; under (K) the chromatic number is correct but the diameter is not. Every other reading fails by Corollary 4. For $n = 2, 3$, `verify.py` also checks all 7 and 127 Cayley graphs $\text{Cay}((\mathbb{Z}/2)^n, S)$ ($S \neq \emptyset$) directly. For $N = 4$ it checks all 64 labelled graphs: the only one with $\chi = 4$ is K_4 , and it has diameter 1.

5 Robustness remarks

Remark 6 (Disconnected graphs). If $Q_{\oplus}(n)$ is disconnected its diameter is $\infty \neq 2^n - 1$, so the claim fails anyway. Theorem 3 already covers this: $\chi = N$ forces completeness, hence connectivity. In the formalization, the diameter predicate requires every pair to be joined by a walk of length at most D , so a disconnected graph has no finite diameter.

Remark 7 (A different vertex count). The conjecture says the graph lives on the hypercube, so it has 2^n vertices. For a reader who would allow other vertex sets, here is the general inequality: every connected graph on N vertices satisfies

$$\chi(G) + \text{diam } G \leq N + 1.$$

Indeed, let $d = \text{diam } G \geq 1$ and let $v_0 v_1 \dots v_d$ be a shortest path between two vertices at distance d . If $|i - j| \geq 2$ then v_i and v_j are not adjacent, since otherwise there would be a shorter path. So colour the v_i with even i by colour 1, those with odd i by colour 2, and the remaining $N - d - 1$ vertices with distinct new colours. This gives $\chi \leq N - d + 1$. The case $d = 0$, i.e. $N = 1$, is trivial. Hence $\text{diam} = 2^n - 1$ and $\chi = 2^n$ would need $N \geq 2^{n+1} - 2$ vertices. For example, the clause fails at $n = 2$ even on 5 vertices. The bound is tight: K_{2^n} with a pendant path of length $2^n - 2$ attached to one of its vertices has $2^{n+1} - 2$ vertices, chromatic number 2^n , and diameter $1 + (2^n - 2) = 2^n - 1$ (from a clique vertex other than the attachment point to the far end of the path). This does not affect the conjecture, which fixes the vertex set to the 2^n vertices of the hypercube; there the clause is impossible by Corollary 4. `verify.py` confirms the inequality for all $2^{15} + 2^{10} + 2^6 + 2^3$ labelled graphs on $N = 3, \dots, 6$ vertices. This remark is not formalized in Lean, and it is not needed for the disproof.

Remark 8 (Directed readings). Edges “given by $\oplus$ ” are undirected, because $u \oplus v = v \oplus u$. Suppose one nevertheless reads $Q_{\oplus}(n)$ as a game digraph (“nim” moves), with χ taken from the underlying undirected graph and the diameter measured along directed paths. Then $\chi = N$ still forces the underlying graph to be complete, by Lemma 2. The acyclic digraphs whose underlying graph is complete are exactly the transitive tournaments: there is one arc between each pair, and the arcs follow a linear order. In such a digraph every reachable ordered pair is joined by a single arc, so all reachable distances equal 1. The reverse pairs are unreachable, so the directed diameter is ∞ . In neither case is it $2^n - 1$. A cyclic, non-symmetric arc rule can reach directed diameter 3 on 4 vertices, and `verify.py` finds 192 such digraphs over K_4 , none symmetric. Such digraphs exist for every $N \geq 3$. For example, take the arcs $i \rightarrow i + 1$ and all backward arcs $j \rightarrow i$ with $j > i + 1$: the underlying graph is K_N , and the directed diameter is $N - 1$. But no rule depending on $u \oplus v$ produces one, because $\oplus$ is commutative, so such a rule is not a reading of the conjecture. This remark is likewise only in the report and in `verify.py`.

Remark 9 (Grundy chromatic number). Suppose χ is read as the Grundy (first-fit) chromatic number Γ : the largest number of colours used by greedy colouring over all vertex orderings. In a Grundy colouring, every vertex of colour k is adjacent to vertices of every colour $< k$. If $\Gamma = N$, the N colour classes are singletons, so the vertex of each colour is adjacent to the vertex of every smaller colour. Then $G = K_N$ again, with diameter 1, so this reading fails too. This remark is checked by `verify.py` for $N = 4$ and is not formalized in Lean.

6 Lean formalization

The directory `lean4/` is a self-contained Lean 4.19.0 project (core library only, no Mathlib). All objects are defined from scratch in `Main.lean`.

- `Graph := Nat -> Nat -> Bool`, restricted to $\{0, \dots, N-1\}$; `IsSimple G N` means symmetric and loopless.
- `IsProperColoring`, `Colorable G N c`, and `ChromaticNumberEq G N c`: a proper c -colouring exists and no proper k -colouring exists for $k < c$ (“ c is the least”).
- `WalkFrom G N u ws v` (an explicit list of vertices forming a walk), `WalkLe G N u v k` (a walk of length $\leq k$ exists, i.e. $\text{dist} \leq k$), `DistEq`, and `DiameterEq G N D`: all pairs are at distance $\leq D$ and some pair is at distance exactly D . A Boolean reachability function `reach` is proved equivalent to `WalkLe` (`reach_iff`), and is used only for decidability.
- Well-definedness: `chromatic_unique`, `diameter_unique`, `colorable_self`.
- Key lemma `adj_of_chromatic` (Lemma 2), `diameter_le_one_of_chromatic`, and the general theorem `no_graph_diam_chi`: for all $N \geq 3$ and every simple G on N vertices, $\neg(\text{diam} = N-1 \wedge \chi = N)$.
- The conjecture: `ClaimAt G n` is $\text{diam } G = 2^n - 1 \wedge \chi(G) = 2^n$ on 2^n vertices, and `NimClaim Q` is $\forall n \geq 1, \text{ClaimAt } (Q \oplus n) n$. `claimAt_false`: for every $n \geq 2$ and every simple G on 2^n vertices, $\neg \text{ClaimAt } G n$. Main theorem:

```
theorem conjecture_00000001289_false (Q : Nat -> Graph)
  (hQ : IsSimple (Q 2) (2 ^ 2)) : Not (NimClaim Q)
```

(In `Main.lean` the arrows and the negation are written with the usual Unicode symbols.) This covers every family $Q_\oplus$, with no assumption on its edges beyond $Q_\oplus(2)$ being a simple graph.

- Concrete readings: `hypercube` ($u \oplus v$ a power of two) and `nimComplete` ($u \oplus v \neq 0$), both defined with Lean’s bitwise `^^^`. Proved: `hypercube2_chi`, `hypercube3_chi` ($\chi = 2$), `hypercube2_diam`, `hypercube3_diam` (diameter 2, 3), `nimComplete_chi` ($\chi = N$ for all N , via pigeonhole), `nimComplete_diam` (diameter 1 for $N \geq 2$), and `hypercube_reading_false`, `complete_reading_false`.
- Non-vacuity: `claimAt_one` (the clause holds at $n = 1$ for both readings) and `nonvacuous_four` (on 4 vertices the path has diameter 3 and K_4 has $\chi = 4$, so each conjunct alone is satisfiable).
- Cross-check: `all_masks_fail`. For each of the 64 labelled graphs on $\{0, 1, 2, 3\}$, encoded by an edge bitmask, either a proper 3-colouring is found or all distances are ≤ 2 ; hence none has $\text{diam} = 3$ and $\chi = 4$. This proof uses `decide` and is independent of the general lemma.

`lake build` succeeds with no errors or warnings. There is no `sorry`, no `native_decide`, and no added axioms. `#print axioms` reports only the standard `propext`, `Quot.sound` and, for the general theorems, `Classical.choice`, which comes from tactic-generated case splits.

Scope of the formal result. Lean covers every simple graph on the 2^n vertices $\{0, \dots, 2^n - 1\}$, i.e. every undirected reading of “edges given by $\oplus$ ”. Remarks 7, 8 and 9 (other vertex counts, directed readings, Grundy number) are only in this report and in `verify.py`. The general- n values for the hypercube reading (Proposition 5(H)) are proved here on paper, and in Lean only for $n = 1, 2, 3$. None of these is needed for the disproof.

7 Independent check: `verify.py`

The Python 3 script (standard library only) uses brute force: breadth-first search for distances and backtracking for colourings. It checks: (i) all 64 graphs on 4 vertices, giving the distribution of (χ, diam) , with no graph having $(4, 3)$; (ii) $\chi = N$ only for K_N , and $\chi + \text{diam} \leq N + 1$, for all graphs on $3 \leq N \leq 6$ vertices; (iii) readings (H) and (K) for $n = 1, \dots, 5$, and all Cayley graphs of $(\mathbb{Z}/2)^n$ for $n = 2, 3$; (iv) the directed remark on 4 vertices, and the directed family above for $N = 3, \dots, 8$; (v) the tight example for the vertex-count bound, for $n = 1, 2, 3$; (vi) on 4 vertices, only K_4 has Grundy number 4.

8 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, core only
cd .. && python3 verify.py # Python 3 standard library
pdflatex report.tex && pdflatex report.tex
```